Technical Document #223: Natural Language Processing - Comprehensive Overview

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The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Text summarization generates a concise summary of a longer text. Extractive summarization selects important sentences while abstractive summarization generates new text.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Key Takeaways:

- Machine translation converts text from one language to another.
- Question answering systems extract answers from a given context.
- Sentiment analysis determines the emotional tone behind a body of text.